Noah Hickman

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Objective: Test and validation-focused undergraduate mechanical engineer with hands-on experience at Volvo and Daimler and leadership in FSAE and Solar Car. Seeking to prototype, integrate, and validate dynamic mechanical & fluid systems.

EDUCATION

B.S. Mechanical Engineering – The University of Texas at Austin **May 2027**

- **GPA:** 3.84; **Relevant Coursework:** Heat Transfer, Robot Mechanism Design, MATLAB, Dynamics, Solids
- **Scholarships:** Jim Moritz Memorial Scholarship in Automotive Engineering; Ford Motor Company Scholarship

SKILLS

Programming & Data: Python (pandas, NumPy, matplotlib, scikit-learn, psycopg2), MATLAB, SQL, Docker/Portainer, Git
Simulation & Testing: ANSYS Mechanical (FEA), ATI Vision, Vector CANalyzer, Simulink, Hondata/KTuner
CAD & Manufacturing: SolidWorks, Fusion 360, TIG Welding, 3D Printing, GD&T, Lathe and Manual Mill, Soldering
General: Microsoft Office, Confluence, Jira, Agile, U.S. citizen with no work restrictions, trained in ITAR compliance

WORK EXPERIENCE

Product Testing Mechatronics (PTM) Intern – Daimler Trucks North America **May 2025 – August 2025**

- Built and deployed a standalone Python script package with custom machine learning and localized semantic models to transcribe voice trigger logs, detect optical acoustic warnings (OAWs), and automate test data tagging
- Merged Python tools with the PTM data platform and SQL backend, streamlining analysis across multiple testing teams
- Supported PVE vehicle feature testing (ADAS/Brakes) and data acquisition during Pacific Northwest Over-the-Road tests

Powertrain Integration Engineering Co-Op – Volvo Group **January 2025 – May 2025**

- Performed root cause analysis on diesel powertrain faults and sensor logic using MATLAB and Simulink documentation
- Integrated chassis and powertrain hardware, harnesses, and sensors to validate CARB-compliant engine packages
- Extracted CAN data from vehicle ECMs using ATI Vision and CANalyzer to troubleshoot 56+ engine and exhaust parameters
- Initiated and built Python tools for statistical analysis of powertrain data, reducing manual troubleshooting time by 50%

TECHNICAL ACTIVITIES AND PROJECTS

Longhorn Racing (Formula SAE, Formula Sun Grand Prix, American Solar Challenge)

Combustion Engine Calibration Lead **May 2025 – Present**

- Spearheading event-specific calibration of Honda CBR600RR engine while exploring the use of an E85 fueling system

Solar Mechanical Systems Mentor **January 2025 – May 2025**

- Served as an advisor for design reviews and member training, mentoring over 45 mechanical team members

Solar Body Lead **July 2024 – January 2025**

- Led the 15-member Frame/Ergonomics subsystems, organized meetings, and created a high-level design roadmap
- Reduced ANSYS simulation time by 80% by generating MATLAB scripts to automate stiffness calculations
- Launched team-wide research on structural composites, testing alternatives to conventional frame structures

Solar Frame Sublead **January 2024 – July 2024**

- Utilized SolidWorks CAD and Ansys Mechanical to validate and complete the chassis design based on ASC regulations
- TIG-welded space frame assembly with .001" tolerance mounting tabs, using custom welding fixtures

Solar Frame Subsystem Member **August 2023 – January 2024**

- Modeled tubular frame roll-cage structure and cross-bracing in SolidWorks; supported suspension part manufacturing

Propulsion Subsystem Member – Longhorn Rocketry Association (Spaceport America) **February 2024 – July 2024**

- Designed a converging-diverging nozzle and an impinging injector for the Taurus II hybrid rocket engine using SolidWorks
- Conducted trade studies to determine optimal parts and materials, including nozzle O-rings

Flight-Weight Fluids System Member – Texas Rocket Engineering Lab (Spaceport America) **August 2023 – August 2024**

- Compiled valve/fitting specs from P&ID schematics to meet high-pressure fluid system requirements
- Helped operate a LabQuest DAQ to characterize pressurant flow on an orifice flow bench at Firefly Aerospace test facility

Personal Project Car - 1997 Acura Integra: Planned and integrated multiple fueling, cooling, and drivetrain upgrades; Street-tuned via Hondata S300 ECU module; Goal of designing and building a custom supercharger system for B18B1 engine

Hobbies: Project car building, record/vinyl collecting, and solo traveling; Member of UT ASME